UNIVERSITY OF BIRMINGHAM | GRAVITATIONAL WAVE ASTRONOMY

OBSERVATIONAL COSMOLOGY

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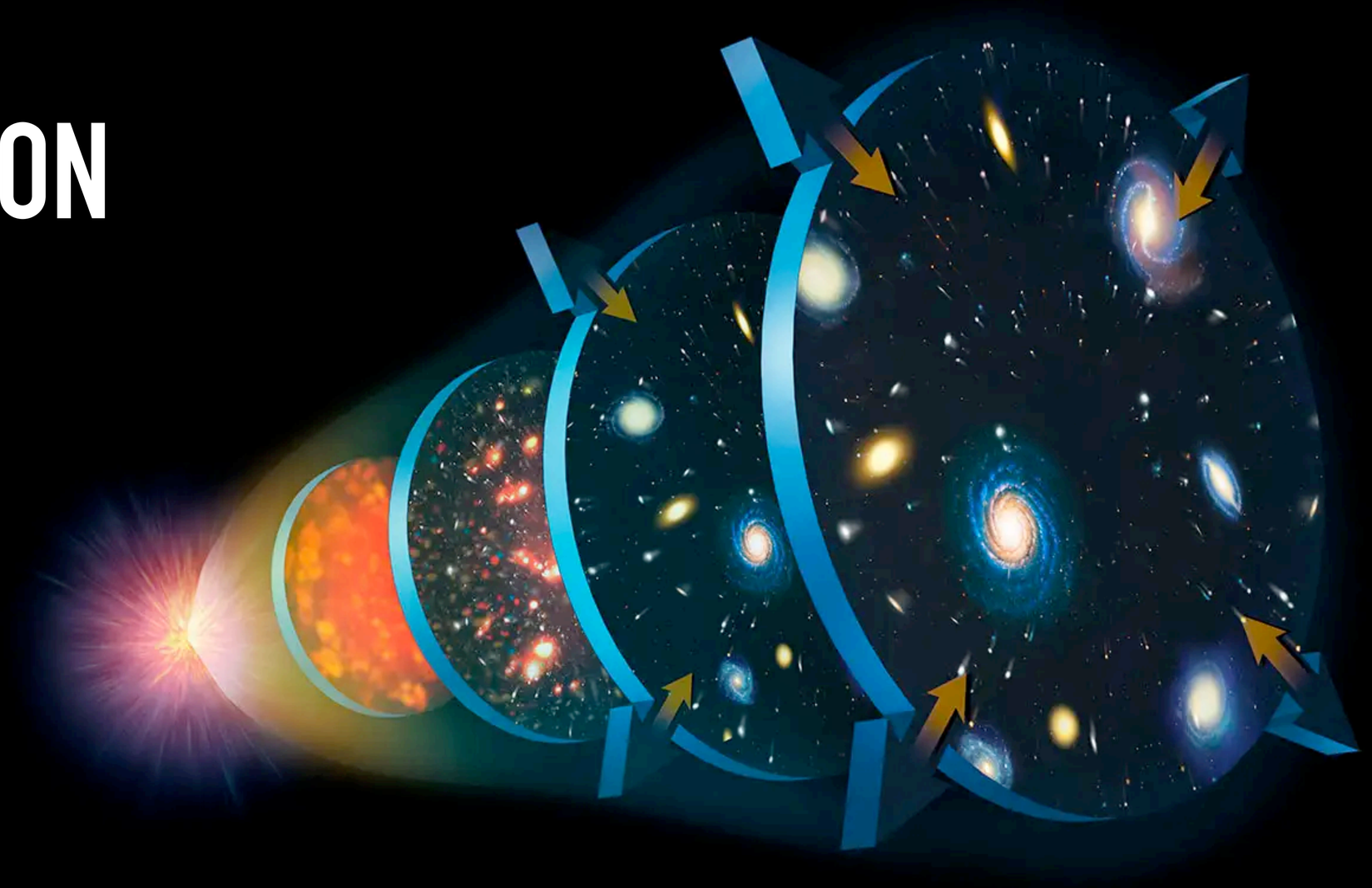
RECAP: BBN

- Deuterium fusion started when $k_B T < 2.22\,\mathrm{MeV}$ corresponding to
  - A temperature of $T_{\mathrm{nuc}} \approx 7.6 \times 10^8\,\mathrm{K}$
  - At a time $t_{\mathrm{nuc}} \approx 200\,s$
- Subsequent fusion of tritium, helium-3, helium-4 (the roadblock) and small amount of lithium/beryllium
  - If every free neutron bound is bound, then $Y_{\mathrm{max}} = 1/3$.
  - At $t \approx 1000\,s$, BBN is effectively over
- BBN yields depend sensitively on the baryon-to-photon ratio η
  - Quasar observation to measure the primordial abundance of deuterium
  - BBN measurement consistent with CMB measurement of η
- Particle asymmetries:
  - Photons vs. baryons
  - Matter vs. antimatter
  - Origin unknown

INFLATION

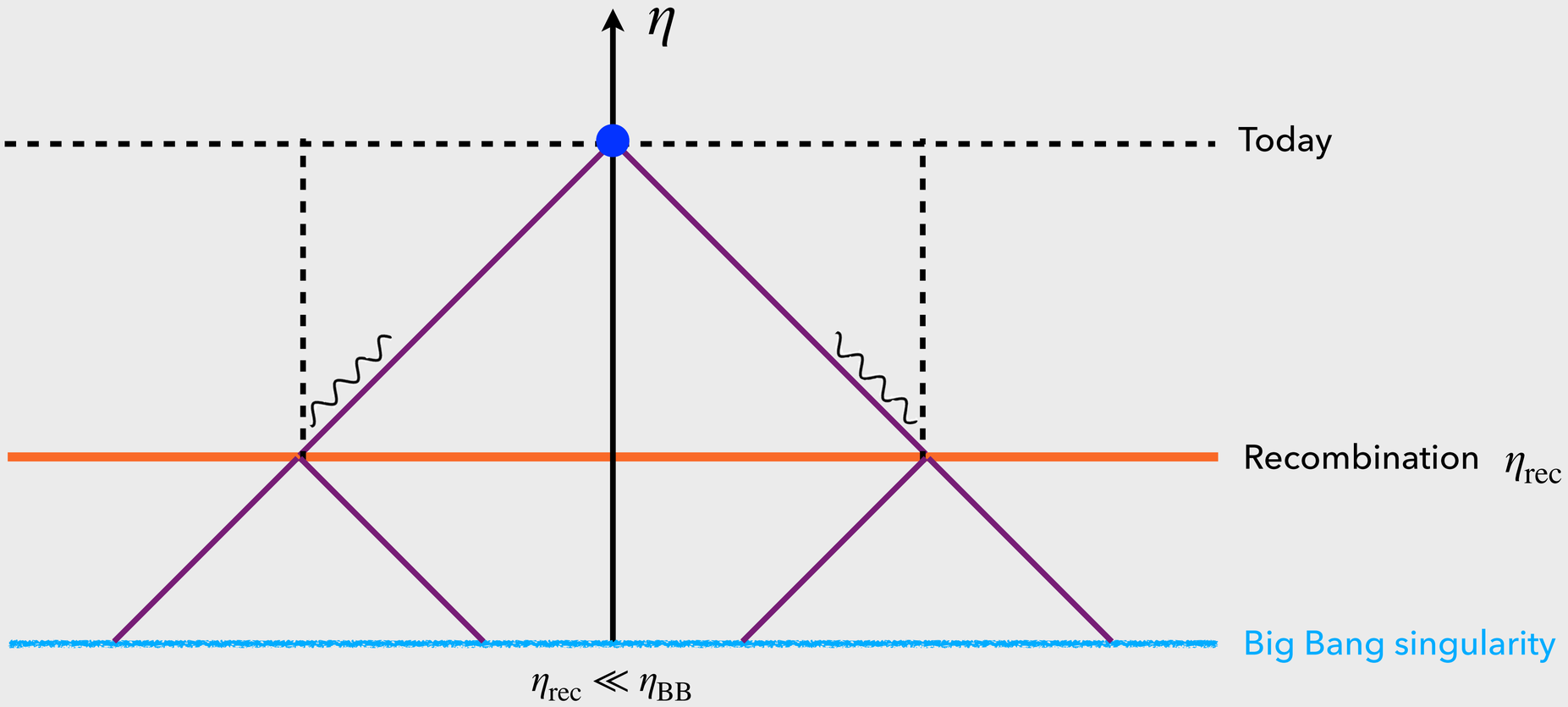


THE HORIZON PROBLEM

- Statement that the universe is **nearly homogeneous and isotropic** on very large scales (reflected in the CMB)
- All events that we can observe are in our past light cone
- Surface of last scattering is the intersection between our past light cone and the hypersurface $\eta = \eta_{\rm rec}$

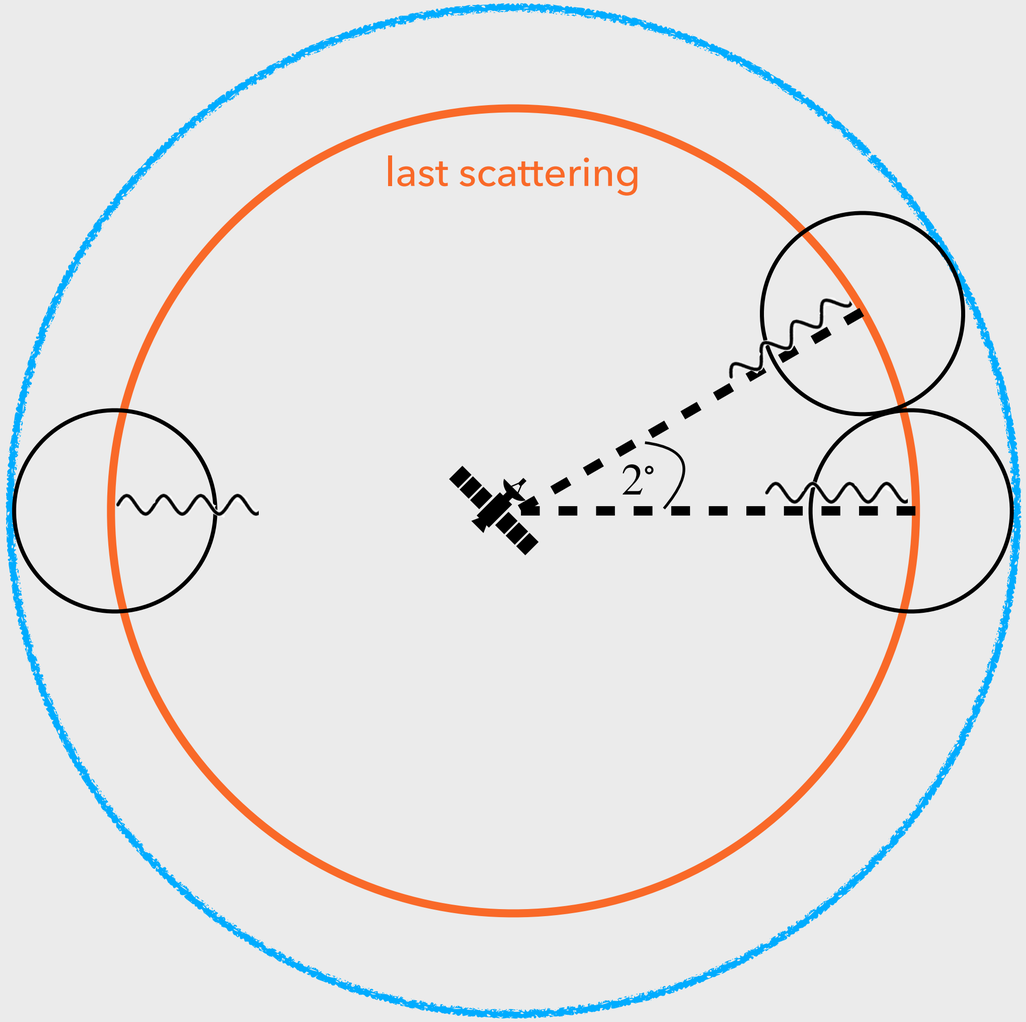
Conformal time:

$$\eta = \int \frac{dt}{a(t)}$$



THE HORIZON PROBLEM

- Most parts of the CMB have **non-overlapping past lightcones**
  - One can show that **any two points** in the CMB separated by more than 2 degrees on the sky were never in causal contact
  - How do two photons from **causally disconnected** regions "know" that they should have almost exactly the same temperature?
  - The homogeneity of the CMB spans scales much larger than the particle horizon but there was not enough time between the initial singularity and last scattering for those patches to communicate.



THE FLATNESS PROBLEM

- The observed value of the total density today points to $|\Omega_{k,0}| < 0.005$ (global spatial flatness)
- Let us extrapolate $\Omega(t)$ back in time:
  - Starting from the Friedmann equation we find

$$1 - \Omega(t) = \Omega_k(t) = \frac{H_0^2(1-\Omega_0)}{a^2(t)H^2(t)}$$

  - At early times, radiation and matter were the dominant components:

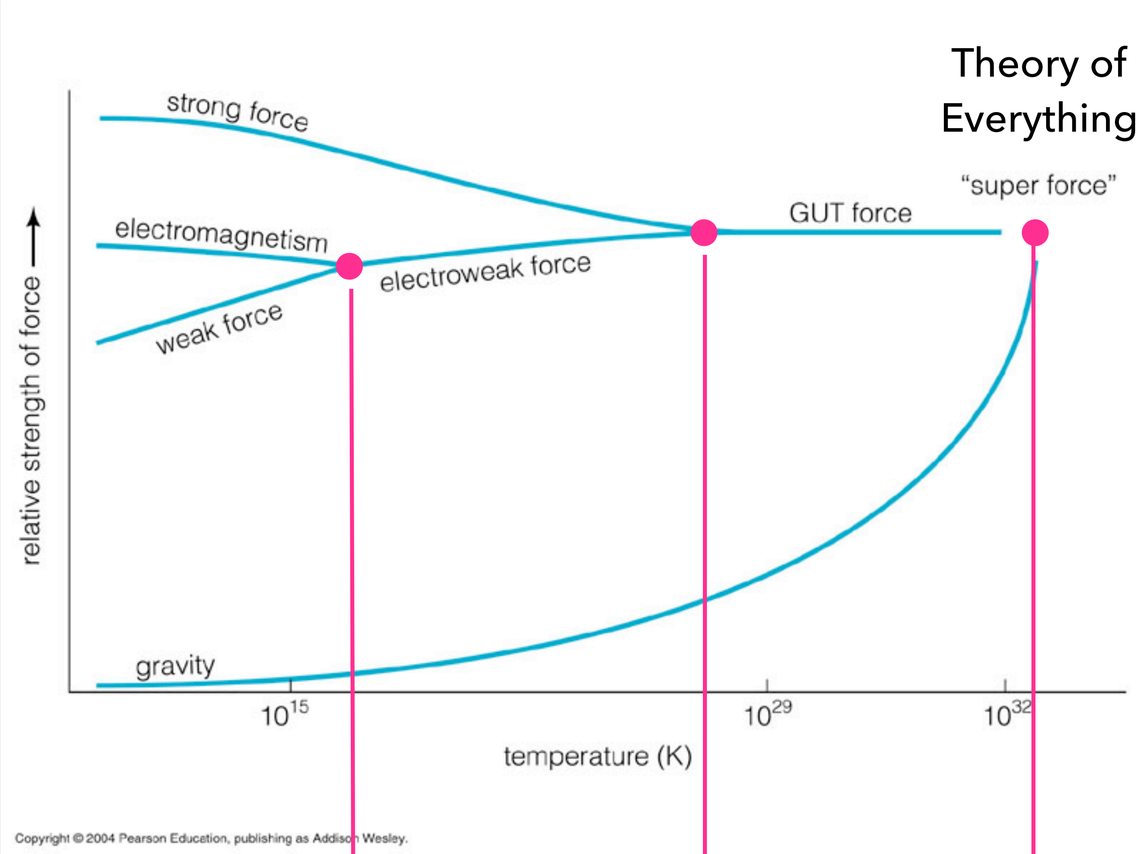
$$\frac{H^2(t)}{H_0^2} = \frac{\Omega_{r,0}}{a^4(t)} + \frac{\Omega_{m,0}}{a^3(t)} \quad \Rightarrow \quad 1 - \Omega(t) = \frac{(1-\Omega_0)a^2(t)}{a(t)\Omega_{m,0} + \Omega_{r,0}}$$

$$|1 - \Omega(t_{\rm eq})| \lesssim 10^{-6}$$
$$|1 - \Omega(t_{\rm nuc})| \lesssim 10^{-16}$$
$$|1 - \Omega(t_{\rm P})| \lesssim 10^{-62}$$

The deviation from 1 was constantly growing. The universe was even flatter at earlier times!

THE MAGNETIC MONOPOLE PROBLEM

- **Where have all the magnetic monopoles gone?**
- **Grand Unified Theory** (GUT):
  - Attempts to unify EM, weak & strong forces
- Predictions: **Phase transitions** as $T < T_{\rm GUT}$
  - Associated with a spontaneous loss of symmetry
  - Gives rise to **topological defects** that act as magnetic monopoles
  - Predicted number density at creation: $n_M(t_{\rm GUT}) \sim 10^{82}\,{\rm m}^{-3}$
  - Predicted energy density: $\epsilon(t_{\rm GUT}) \sim 10^{104}\,{\rm TeV\,m}^{-3}$
  - They would dominate the non-relativistic component today!
  - BUT: There is no observational evidence for magnetic monopoles!



$T \sim 10^{16}{\rm K}$ $E \sim 1\,{\rm TeV}$ $t \sim 10^{-12}s$

$T \sim 10^{28}{\rm K}$ $E \sim 10^{12}\,{\rm TeV}$ $t \sim 10^{-36}s$

$T \sim 10^{32}{\rm K}$ $E \sim 10^{16}\,{\rm TeV}$ $t \sim 10^{-43}s$

COSMIC INFLATION

- **Can a single cosmological mechanism cure those problems?**
  - **Cosmic inflation:** A short (finite) period of **accelerated expansion** (i.e. $\ddot{a} > 0$) in the early universe

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}(\epsilon + 3p)$$

  - For $\ddot{a} > 0$, we require $p < -\epsilon/3$. This can only happen when there is a fluid component with $w < -1/3$ that dominates that phase
    - Alan Guth's original paper: https://journals.aps.org/prd/abstract/10.1103/PhysRevD.23.347

- **Simplest form:** Consider a positive cosmological constant Λ_i (i.e. $w = -1$)
  - From Friedmann equation we find exponential expansion: $a(t) \propto e^{H_i t}$
  - During inflation the size of the universe increased by N e-folds: $\frac{a(t_f)}{a(t_i)} = e^N, \quad N \equiv H_i(t_f - t_i)$
  - Let's assume that $t_i \approx t_{\rm GUT} \approx 10^{-36} s$ and $H_i \approx 1/t_{\rm GUT}$, then $\epsilon_{\Lambda_i} \sim 10^{105}\,{\rm TeV\,m^{-3}}$
    - Comparison: $\epsilon_{\Lambda,0} \approx 10^{-3}\,{\rm TeV\,m^{-3}}$

HOW DOES INFLATION CURE THE PROBLEMS?

- **Flatness problem:**
  - Consider a non-perfectly flat universe, then: $|1-\Omega(t)| = \Omega_k(t)$
  - During inflation, $H(t) = H_i = \mathrm{const}$. and the scale factor grows exponentially, thus

$$|1-\Omega(t)| \propto e^{-2H_i t}$$

  - The deviation from 1 decreases exponentially with time.
  - If we compare the density parameters at the beginning and end of inflation, we find:

$$|1-\Omega(t_f)| = e^{-2N}|1-\Omega(t_i)|$$

  - Today: $|1-\Omega_0| \leq 0.005$. Extrapolating back in time to $t_f = (N+1)t_i$ and assuming the GUT scale,

$$|1-\Omega(t_f)| \leq 2\times 10^{-54}(N+1)$$

  - One can show that $N \approx 60$ **matches your observations** even if the universe was strongly curved prior to the onset of inflation. Inflation is very efficient at flattening the global curvature.

HOW DOES INFLATION CURE THE PROBLEMS?

- **Horizon problem:**
  - Prior to inflation the universe was radiation-dominated
  - (Particle) Horizon distance $d_H(t)$ at the beginning of inflation:

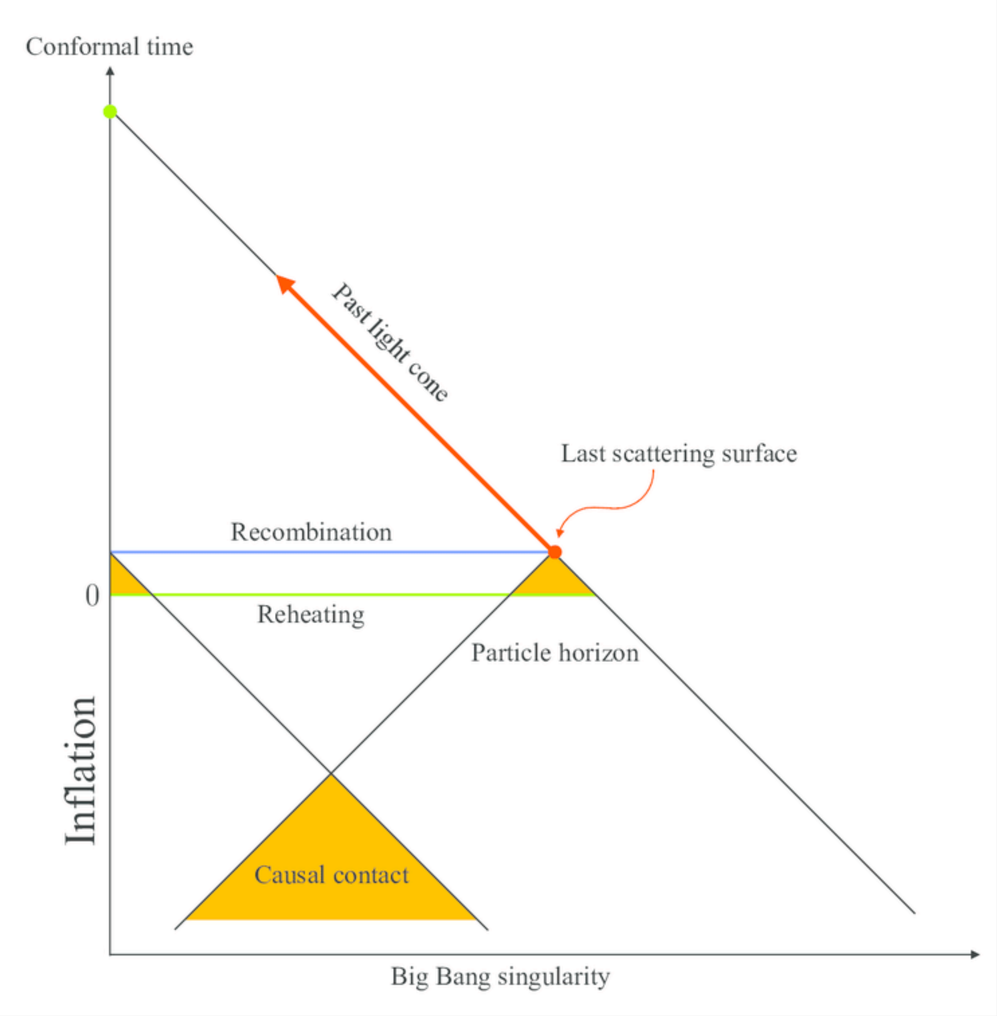
$$d_H(t_i) = 2ct_i$$

  - (Particle) horizon distance at the end of inflation:

$$d_H(t_f) = e^N c(2t_i + H_i^{-1})$$

  - A period of inflation causes the **horizon size to grow exponentially!**
    - Example (GUT scale): $10^{-28}\,\mathrm{m} \rightarrow \sim 15\,\mathrm{m}$

- **Monopole problem:**
  - Exponential decrease in n_M during inflation



[Baumann, TASI lectures]

PHYSICS OF INFLATION

- What triggered inflation at t_i?
- What switched inflation off at t_f?
- Why does it not reduce the number of photons to undetectable levels?
- Why does inflation not flatten out local curvature, i.e., density fluctuations that seed the growth of large-scale structure?

- **Simplest inflationary models:**
  - Inflation driven by the evolution of a **single scalar field** $\phi(t, \vec{x})$ called **inflaton** with dimension energy
  - Associated **potential energy** $V(\phi)$ of dimension energy density
    - If the field is dynamical, it also carries a **kinetic energy**